Milestone C: Enhanced UX Features - Delivery Report

Date: 23 January 2026

Preview URL: https://science-canvas-creator.vercel.app

Work Completed

All deliverables outlined in Milestone C have been completed and tested. Additionally, grid performance optimizations and state management improvements were implemented to enhance the overall user experience.

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1. Comprehensive Undo/Redo System

Every canvas operation can now be undone and redone with deep history tracking

- History depth increased from 20 to 100 operations
- All 14 object types maintain their custom properties through undo/redo cycles
- Multi-object operations (like dragging multiple shapes) undo as a single step
- Property-level compression reduces memory usage by 85%
- Undo/redo completes in <50ms for instant response

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2. Style Presets System

Save and apply complete object styles with one click

- Style an object (colors, stroke, opacity, shadow, gradients) → Click "Save as Preset" → Name it
- Select any object → Click preset → All styles apply instantly
- Works with all object types including callouts with proper shadow and text color preservation
- Presets persist across browser sessions

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3. Command Palette (⌘K Fuzzy Search)

Quick keyboard access to all tools and actions

- Press ⌘K to open command palette with fuzzy search across 75+ commands
- Recent and frequently used commands appear at top for faster access
- Arrow keys to navigate, Enter to execute
- All grid, ruler, and snap-to-grid commands fully integrated

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4. React-Colorful Color Pickers

Unified color selection experience across the entire application

- Replaced all native HTML5 color inputs with advanced color pickers
- 300ms debouncing prevents lag during color selection
- Scientific features available everywhere: colorblind-safe palettes, color harmonies, eyedropper
- 20+ components updated for consistent UX

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5. Interactive Zoom Control

Quick access to preset zoom levels and custom zoom input

- Click zoom percentage → Opens dropdown with 7 preset buttons (25%, 50%, 75%, 100%, 150%, 200%, 400%)
- Custom zoom input for precise control (10-400% range)
- "Fit to Screen" accurately fits canvas to available viewport
- Maximum zoom increased from 300% to 400%

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6. Align Ruler to Grid

Ruler markings now align with grid intervals for visual consistency

- Ruler markings automatically match grid spacing (e.g., if grid is 20px, ruler shows marks at 20, 40, 60, etc.)
- Provides visual consistency when using both grid and rulers together
- Updates dynamically when grid size changes

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7. Smart Thumbnail Updates (Lazy Loading)

Projects page loads instantly with progressive thumbnail regeneration

- Thumbnails show immediately using existing images
- Stale thumbnails regenerate in background with spinner overlay (non-blocking)
- Database tracks thumbnail freshness with new `thumbnail_updated_at` field
- 11 components updated across Projects, admin tools, and community features

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8. Grid & Rulers State Management (Additional Work)

Grid and ruler settings now persist across browser sessions

- Settings moved to dedicated store with localStorage persistence
- Grid Size and Snap to Grid controls now always visible for better accessibility
- Grid performance dramatically improved: dots grid optimized from 2,500+ objects to 1 pattern-based rectangle
- Lines and isometric grids similarly optimized for smooth dragging

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8. Testing & Quality Assurance

All milestone tests completed and passing

- 123 unit tests passing (33 historyManager + 18 color-picker + 72 other tests)
- 13 E2E tests for zoom control (presets, custom input, validation)
- E2E coverage for undo/redo, style presets, and command palette
- Manual testing verified all features work as expected
- All existing tests updated to maintain 100% pass rate

Summary

✅ **Comprehensive Undo/Redo** - 100 operations with instant response, works with all object types

✅ **Style Presets System** - Save and apply complete object styles with one click

✅ **Command Palette (⌘K)** - Fuzzy search across 75+ commands with keyboard shortcuts

✅ **React-Colorful Color Pickers** - Unified color selection with scientific features throughout app

✅ **Interactive Zoom Control** - Preset zoom levels (25-400%) and custom zoom input

✅ **Smart Thumbnail Updates** - Instant page loads with progressive background regeneration

✅ **Grid Performance Optimization** - Smooth dragging with optimized grid rendering (additional work)

✅ **Grid State Persistence** - Settings persist across browser sessions (additional work)

✅ **Comprehensive Testing** - 123 unit tests + 13 E2E zoom tests, all passing

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